Phys. 2110 HW # 15 15.12, 16, 30, 51, 61, 64, 73

15.12 (a) $A\cos 2\pi\left(\frac{x}{\lambda}-\frac{t}{T}\right) = A\cos\frac{2\pi}{\lambda}\left(x-\frac{\lambda}{T}t\right) = A\cos\frac{2\pi}{\lambda}(x-vt)$

(b) $V_y = \frac{\partial y}{\partial t} = \frac{2\pi V}{\lambda}A\sin\frac{2\pi}{\lambda}(x-vt)$

(c) The speed is the greatest when the sine is 1, and that speed is $\frac{2\pi VA}{\lambda}$. This will be equal to V if $A=\frac{\lambda}{2\pi}$, less than V if $A<\frac{\lambda}{2\pi}$ and greater than V if $A>\frac{\lambda}{2\pi}$.

15.16 Comparison to the general equation gives $A = 8.50\,mm$, $k = 172\,rad/m$ and $\omega = 2730\,rad/s$. The string has mass 0.128 kg and $\mu = \frac{m}{L} = 0.0850\,kg/m$

(a) $V = \frac{\omega}{k} = 15.9\,m/s$, $t = \frac{d}{V} = \frac{1.50\,m}{15.9\,m/s} = 0.0943\,s$.

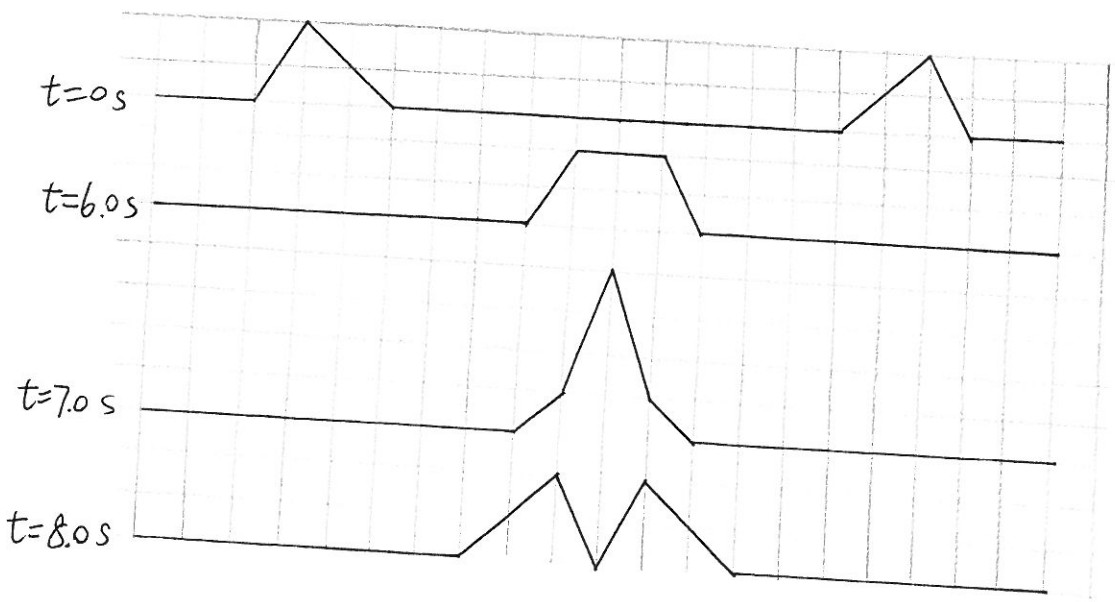
(b) $W = F = \mu V^2 = (0.0850\,kg/m)(15.9\,m/s)^2 = 21.5\,N$

(c) $\lambda = \frac{2\pi}{k} = \frac{2\pi\ rad}{172\ rad/m} = 0.0365\,m$

The number of wavelengths along the length of the string is $\frac{1.50\,m}{0.0365\,m} = 41.1$

(d) For a wave traveling in the opposite direction,
$y(x,t) = (8.50\,mm)\cos(172\,m^{-1}x + 2730\,s^{-1}t)$

15.30



15.51 (a) $v = \sqrt{\frac{F}{\mu}}$ $\qquad t = \frac{L}{v}$

The travel times for each segment are $t_1 = L\sqrt{\frac{\mu_1}{F}}$, $t_2 = L\sqrt{\frac{4\mu_1}{F}}$, and $t_3 = L\sqrt{\frac{\mu_1}{4F}}$

The total time is $t_{total} = L\sqrt{\frac{\mu_1}{F}} + 2L\sqrt{\frac{\mu_1}{F}} + \frac{1}{2}L\sqrt{\frac{\mu_1}{F}} = \frac{7}{2}L\sqrt{\frac{\mu_1}{F}}$

(b) No. The speed in a segment depends only on F and μ for that segment.

15.61 (a) As time goes on, someone moving with the wave would need to move in such a way that the wave appears to have the same shape. If this motion can be described by $x = vt + b$, with b a constant, then $y(x,t) = f(b)$, and the waveform is the same to such an observer.

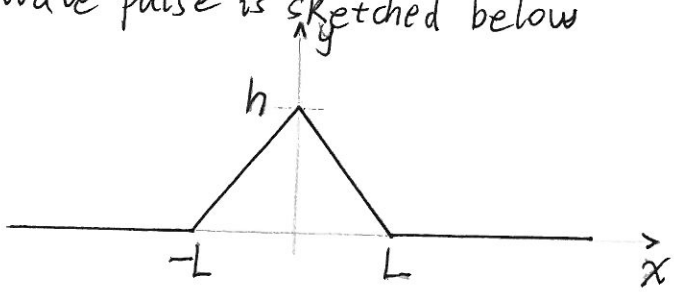
(b) $\frac{\partial^2 y}{\partial x^2} = \frac{d^2 f}{du^2}$ and $\frac{\partial^2 y}{\partial t^2} = v^2 \frac{d^2 f}{du^2}$ $\qquad \frac{\partial^2 y}{\partial x^2} = \frac{1}{v^2}\frac{\partial^2 y}{\partial t^2}$

$y(x,t) = f(x - vt)$ is a solution to the wave function with speed v.

(c) $y(x,t) = De^{-(Bx - Ct)^2} = De^{-B^2(x - \frac{C}{B}t)^2}$

The speed of this wave $v = \frac{C}{B}$

15.64 (a) The wave pulse is sketched below



(b)

$$y(x,t) = \begin{cases} 0 & \text{for } (x - vt) < -L \\ h(L + x - vt)/L & \text{for } -L < (x - vt) < 0 \\ h(L - x + vt)/L & \text{for } 0 < (x - vt) < L \\ 0 & \text{for } (x - vt) > L \end{cases}$$

(c) From Eq. (15.21)

$$P(x,t) = -F\frac{\partial y(x,t)}{\partial x}\frac{\partial y(x,t)}{\partial t} = \begin{cases} -F(0)(0) = 0 & \text{for } (x - vt) < -L \\ -F(h/L)(-hv/L) = Fv(h/L)^2 & \text{for } -L < (x - vt) < 0 \\ -F(-h/L)(hv/L) = Fv(h/L)^2 & \text{for } 0 < (x - vt) < L \\ -F(0)(0) = 0 & \text{for } (x - vt) > L \end{cases}$$

Thus the instantaneous power is zero except for $-L < (x - vt) < L$, where it has constant value $Fv(h/L)^2$.

15.73 (a) For reflection from a free end of a string the reflected wave is not inverted, so $y(x,t) = y_1(x,t) + y_2(x,t)$, where

$y_1(x,t) = A\cos(kx + \omega t)$ (traveling to the left)

$y_2(x,t) = A\cos(kx - \omega t)$ (traveling to the right)

Apply $\cos(a \pm b) = \cos a \cos b \mp \sin a \sin b$

Then $y(x,t) = (2A\cos kx)\cos\omega t$

(b) For $x=0$, $\cos kx = 1$ and $y(x,t) = 2A\cos\omega t$. The amplitude of the simple harmonic motion at $x=0$ is $2A$, which is the maximum for this standing wave, so $x=0$ is an antinode.

(c) $y_{max} = 2A$ from part (b)

$$V_y = \frac{\partial y}{\partial t} = \frac{\partial}{\partial t}[(2A\cos kx)\cos\omega t] = -2A\omega\cos kx \sin\omega t$$

At $x=0$, $V_y = -2A\omega\sin\omega t$ and $(V_y)_{max} = 2A\omega$

$$a_y = \frac{\partial^2 y}{\partial t^2} = -2A\omega^2\cos kx\cos\omega t$$

At $x=0$, $a_y = -2A\omega^2\cos\omega t$ and $(a_y)_{max} = 2A\omega^2$.